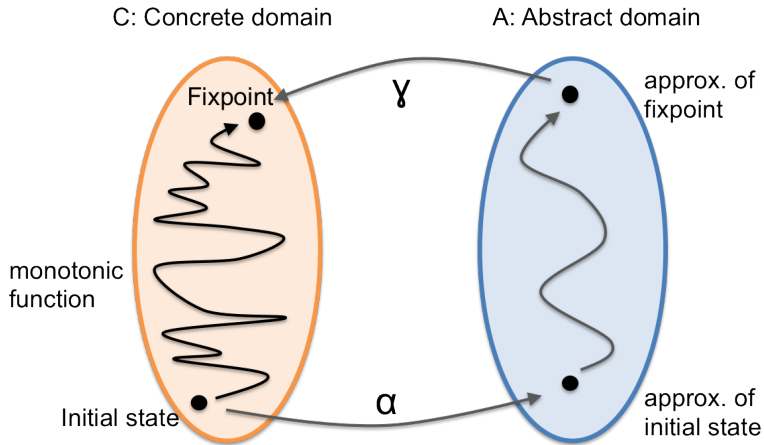


Abstract Interpretation Recipe

Viktor Kunčák

Abstract Interpretation Big Picture



Galois Connection

Galois connection (named after Évariste Galois) is defined by two monotonic functions $\alpha : C \rightarrow A$ and $\gamma : A \rightarrow C$ between partial orders $\leq$ on C and $\sqsubseteq$ on A , such that

$$\forall c, a. \quad \alpha(c) \sqsubseteq a \iff c \leq \gamma(a) \quad (*)$$

(intuitively the condition means that c is approximated by a).

Lemma: The condition $(*)$ holds iff the conjunction of these two conditions:

$$\begin{array}{l} c \leq \gamma(\alpha(c)) \\ \alpha(\gamma(a)) \sqsubseteq a \end{array}$$

holds for all c and a .

Abstract Interpretation Recipe: Setup

Given control-flow graph: (V, E, r) where

- ▶ $V = \{v_1, \dots, v_n\}$ is set of program points
- ▶ $E \subseteq V \times V$ are control-flow graph edges
- ▶ $r : E \rightarrow 2^{S \times S}$, so each $r(v, v') \subseteq S \times S$ is relation describing the meaning of command between v and v'

Key steps:

- ▶ design abstract domain A that represents sets of program states
- ▶ define $\gamma : A \rightarrow C$ giving meaning to elements of A
- ▶ define lattice ordering $\sqsubseteq$ on A such that $a_1 \sqsubseteq a_2 \rightarrow \gamma(a_1) \subseteq \gamma(a_2)$
- ▶ define $sp^\# : A \times 2^{S \times S} \rightarrow A$ that maps an abstract element and a CFG statement to new abstract element, such that $sp(\gamma(a), r) \subseteq \gamma(sp^\#(a, r))$

For example, by defining function α so that (α, γ) becomes a *Galois Connection* and defining $sp^\#(a) = \alpha(sp(\gamma(a), r))$.

Running Abstract Interpretation

- ▶ Extend $sp^\#$ to work on control-flow graphs, by defining $F^\# : (V \rightarrow A) \rightarrow (V \rightarrow A)$ as follows (below, $g^\# : V \rightarrow A$)

$$F^\#(g^\#)(v') = \text{Init}(v') \sqcup \bigsqcup_{(v,v') \in E} sp^\#(g^\#(v), r(v, v'))$$

- ▶ Compute $g_*^\# = \text{lfp}(F^\#)$ (this is easier than computing semantics because lattice A^n is simpler than C^n):

$$g_*^\# = \bigsqcup_{n \geq 0} (F^\#)^n(\perp^\#)$$

where $\perp^\#(v) = \perp_A$ for all $v \in V$.

The resulting fixpoint describes an inductive program invariant.

Concrete Domain: Sets of States

Because there is only one variable:

- ▶ state is an element of $\mathbb{Z}$ (value of x)
- ▶ sets of states are sets of integers, $C = 2^{\mathbb{Z}}$ (concrete domain)
- ▶ for each command K , strongest postcondition function $sp(\cdot, K) : C \rightarrow C$

Strongest Postcondition

Compute sp on example statements:

$$sp(P, x := 0) = \text{if } P = \emptyset \text{ then } \emptyset \text{ else } \{0\}$$

$$sp(P, x := x + 3) = \{x + 3 \mid x \in P\}$$

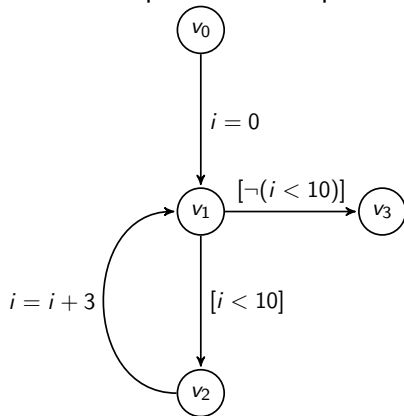
$$sp(P, \text{assume}(x < 10)) = \{x \mid x \in P \wedge x < 10\}$$

$$sp(P, \text{assume}(\neg(x < 10))) = \{x \mid x \in P \wedge x \geq 10\}$$

Programs as control-flow graphs

One possible corresponding control-flow graph is:

```
// v0  
i = 0;  
while // v1  
  (i < 10) {  
    // v2  
    i = i + 3;  
  }  
// v3
```



Sets of States at Each Program Point

Collecting semantics computes with sets of states at each program point

$$g : \{v_0, v_1, v_2, v_3\} \rightarrow C$$

We sometimes write g_i as a shorthand for $g(v_i)$, for $i \in \{0, 1, 2, 3\}$.

In the initial state the value of variable is arbitrary: $I = \mathbb{Z}$

post Function for the Collecting Semantics

From here we can derive F that maps g to new value of g :

$$\begin{aligned} F(g_0, g_1, g_2, g_3) = & \\ & (\mathbb{Z}, \\ & sp(g_0, x := 0) \cup sp(g_2, x := x + 3), \\ & sp(g_1, assume(x < 10)), \\ & sp(g_1, assume(\neg(x < 10)))) \end{aligned}$$

Sets of States at Each Program Point

The fixpoint condition $F(g) = g$ becomes a system of equations

$$g_0 = \mathbb{Z}$$

$$g_1 = sp(g_0, x := 0) \cup sp(g_2, x := x + 3)$$

$$g_2 = sp(g_1, assume(x < 10))$$

$$g_3 = sp(g_1, assume(\neg(x < 10)))$$

whereas the postfix point (see Tarski's fixpoint theorem) becomes

$$\mathbb{Z} \subseteq g_0$$

$$sp(g_0, x := 0) \cup sp(g_2, x := x + 3) \subseteq g_1$$

$$sp(g_1, assume(x < 10)) \subseteq g_2$$

$$sp(g_1, assume(\neg(x < 10))) \subseteq g_3$$

Computing Fixpoint

To find the fixpoint, we compute the sequence $F^n(\emptyset, \emptyset, \emptyset, \emptyset)$ for $n \geq 0$:

$(\emptyset, \emptyset, \emptyset, \emptyset)$
 $(\mathbb{Z}, \emptyset, \emptyset, \emptyset)$
 $(\mathbb{Z}, \{0\}, \emptyset, \emptyset)$
 $(\mathbb{Z}, \{0\}, \{0\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3\}, \{0\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3\}, \{0, 3\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3, 6\}, \{0, 3\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3, 6\}, \{0, 3, 6\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3, 6, 9\}, \{0, 3, 6, 9\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \emptyset)$
 $(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \{12\})$
 $(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \{12\})$

Thus, all subsequent values remain the same and $(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \{12\})$ is the fixpoint of collecting semantics equations. In general we may need infinitely many iterations to converge.

Question

Suppose that we have a program that terminates for every possible initial state. Can we always find a finite constant n such that

$$F^n(\emptyset, \dots, \emptyset) = F^{n+1}(\emptyset, \dots, \emptyset)$$

i.e. the sequence such as the one above is guaranteed to stabilize?

Example: Assume an arbitrary initial value and consider the loop. Compute a sequence of sets of states at the point after the increment statement in the loop, following the equations for collecting semantics.

```
if (y > 0) {  
  x = 0  
  while (x < y) {  
    x = x + 1  
  }  
}
```

What always works from omega continuity:

$$lfp(F) = \bigcup_{n \geq 0} F^n(\emptyset, \dots, \emptyset)$$

where $\bigcup$ on a tuple above means taking union of each component separately, so $(A, B) \cup (A', B') = (A \cup A', B \cup B')$.

Variable Range Analysis for Example Program

The general form of abstract interpretation of the collecting semantics is analogous to collecting semantics, but replaces operations on sets with operations on the lattice:

$$F^\# : (V \rightarrow A) \rightarrow (V \rightarrow A)$$

$$F^\#(g^\#)(v') = g_{init}^\#(v') \sqcup \bigsqcup_{(v,v') \in E} sp^\#(g^\#(v), r(v, v'))$$

Here $g_{init}^\#(v')$ will be $\perp$ except at the entry into our control-flow graph, where it approximates the set of initial states at the entry point.

Abstract Analysis Domain

Before we had representation for all possible sets of states:

$$C = 2^{\mathbb{Z}}$$

Here we have representation of only certain states, namely intervals:

$$\begin{aligned} A = & \{\perp\} \cup \\ & \{(-\infty, q] \mid q \in \mathbb{Z}\} \cup \\ & \{[p, +\infty) \mid p \in \mathbb{Z}\} \cup \\ & \{[p, q] \mid p \leq q\} \cup \\ & \{\top\} \end{aligned}$$

Abstract Analysis Domain

The meaning of domain elements is given by a monotonic *concretization function* $\gamma : A \rightarrow C$:

$$\begin{aligned}\gamma(\perp) &= \emptyset \\ \gamma((-\infty, q]) &= \{x \mid x \leq q\} \\ \gamma([p, +\infty)) &= \{x \mid p \leq x\} \\ \gamma([p, q]) &= \{x \mid p \leq x \wedge x \leq q\} \\ \gamma(\top) &= \mathbb{Z}\end{aligned}$$

From monotonicity and $a_1 \sqsubseteq a_1 \sqcup a_2$ it follows

$$\gamma(a_1) \subseteq \gamma(a_1 \sqcup a_2)$$

and thus

$$\gamma(a_1) \cup \gamma(a_2) \subseteq \gamma(a_1 \sqcup a_2)$$

We try to define γ to be as small as possible while satisfying this condition.

Abstract Analysis Domain

Define *abstraction function* $\alpha : C \rightarrow A$ such that

- ▶ $\alpha(s) = [\min s, \max s]$ if those values exist (set is bounded from below and above)
- ▶ $\alpha(s) = [\min s, +\infty)$ if there is lower but no upper bound
- ▶ $\alpha(s) = (-\infty, \max s]$ if there is upper but no lower bound
- ▶ $\alpha(s) = \top$ if there is no upper and no lower bound
- ▶ $\alpha(\emptyset) = \perp$

Lemma: The pair (α, γ) form a Galois Connection.

Abstract Analysis Domain

By property of Galois Connection, the condition $\gamma(a_1) \cup \gamma(a_2) \subseteq \gamma(a_1 \sqcup a_2)$ is equivalent to

$$\alpha(\gamma(a_1) \cup \gamma(a_2)) \sqsubseteq a_1 \sqcup a_2$$

To make $a_1 \sqcup a_2$ as small as possible, we let the equality hold, defining

$$a_1 \sqcup a_2 = \alpha(\gamma(a_1) \cup \gamma(a_2))$$

For example,

$$\begin{aligned} [0, 2] \sqcup [5, 8] &= \alpha(\gamma([0, 2]) \cup \gamma([5, 8])) \\ &= \alpha(\{0, 1, 2, 5, 6, 7, 8\}) \\ &= [0, 8] \end{aligned}$$

Abstract Postcondition

We had: $sp(\cdot, c) : C \rightarrow C$

Now we have: $sp^\#(\cdot, c) : A \rightarrow A$

For correctness, we need that for each $a \in A$ and each command r :

$$sp(\gamma(a), r) \subseteq \gamma(sp^\#(a, r))$$

We would like $sp^\#$ to be *as small as possible so that this condition holds*.

By property of Galois Connection, the condition $sp(\gamma(a), r) \subseteq \gamma(sp^\#(a, r))$ is equivalent to

$$\alpha(sp(\gamma(a), r)) \sqsubseteq sp^\#(a, r)$$

Because we want $sp^\#$ to be as small as possible (to obtain correct result), we let equality hold:

$$sp^\#(a, r) = \alpha(sp(\gamma(a), r))$$

Because we know α, γ, sp , we can compute the value of $sp^\#(a, r)$ by simplifying certain expressions involving sets of states.

Example

For $p \leq q$ we have:

$$\begin{aligned} sp^\#([p, q], x := x + 3) &= \alpha(sp(\gamma([p, q]), x := x + 3)) \\ &= \alpha(sp(\{x \mid p \leq x \wedge x \leq q\}, x := x + 3)) \\ &= \alpha(\{x + 3 \mid p \leq x \wedge x \leq q\}) \\ &= \alpha(\{y \mid p + 3 \leq y \wedge y \leq q + 3\}) \\ &= [p + 3, q + 3] \end{aligned}$$

For K an integer constant and $a \neq \perp$, we have

$$sp^\#(a, x := K) = [K, K]$$

Note that for every command given by relation r , we have

$$\begin{aligned} sp^\#(\perp, r) &= \alpha(sp(\gamma(\perp), r)) \\ &= \alpha(sp(\emptyset, r)) \\ &= \alpha(\emptyset) \\ &= \perp \end{aligned}$$

Abstract Semantic Function for the Program

In Collecting Semantics for Example Program we had

$$\begin{aligned} F(g_0, g_1, g_2, g_3) = \\ (\mathbb{Z}, \\ sp(g_0, x := 0) \cup sp(g_2, x := x + 3), \\ sp(g_1, assume(x < 10)), \\ sp(g_1, assume(\neg(x < 10)))) \end{aligned}$$

Here we have:

$$\begin{aligned} F^\#(g_0^\#, g_1^\#, g_2^\#, g_3^\#) = \\ (\top, \\ sp^\#(g_0^\#, x := 0) \sqcup sp^\#(g_2^\#, x := x + 3), \\ sp^\#(g_1^\#, assume(x < 10)), \\ sp^\#(g_1^\#, assume(\neg(x < 10)))) \end{aligned}$$

Solving Abstract Function

Doing the analysis means computing $(F^\#)^n(\perp, \perp, \perp, \perp)$ for $n \geq 0$:

$(\perp, \perp, \perp, \perp)$
 $(\top, \perp, \perp, \perp)$
 $(\top, [0, 0], \perp, \perp)$
 $(\top, [0, 0], [0, 0], \perp)$
 $(\top, [0, 3], [0, 3], \perp)$
 $(\top, [0, 3], [0, 3], \perp)$
 $(\top, [0, 6], [0, 3], \perp)$
 $(\top, [0, 6], [0, 6], \perp)$
 $(\top, [0, 9], [0, 9], \perp)$
 $(\top, [0, 12], [0, 9], \perp)$
 $(\top, [0, 12], [0, 9], [10, 12])$
 $(\top, [0, 12], [0, 9], [10, 12])$
 $\dots$

Note the approximation (especially in the last step) compared to the collecting semantics we have computed before for our example program.

Soundness: Abstraction Approximates Exact Computation

$F^n(\emptyset, \emptyset, \emptyset, \emptyset) :$

$(\emptyset, \emptyset, \emptyset, \emptyset)$

$(\mathbb{Z}, \emptyset, \emptyset, \emptyset)$

$(\mathbb{Z}, \{0\}, \emptyset, \emptyset)$

$(\mathbb{Z}, \{0\}, \{0\}, \emptyset)$

$(\mathbb{Z}, \{0, 3\}, \{0\}, \emptyset)$

$(\mathbb{Z}, \{0, 3\}, \{0, 3\}, \emptyset)$

$(\mathbb{Z}, \{0, 3, 6\}, \{0, 3\}, \emptyset)$

$(\mathbb{Z}, \{0, 3, 6\}, \{0, 3, 6\}, \emptyset)$

$(\mathbb{Z}, \{0, 3, 6, 9\}, \{0, 3, 6, 9\}, \emptyset)$

$(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \emptyset)$

$(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \{12\})$

$(\mathbb{Z}, \{0, 3, 6, 9, 12\}, \{0, 3, 6, 9\}, \{12\})$

$(F^\#)^n(\perp, \perp, \perp, \perp) :$

$(\perp, \perp, \perp, \perp)$

$(\top, \perp, \perp, \perp)$

$(\top, [0, 0], \perp, \perp)$

$(\top, [0, 0], [0, 0], \perp)$

$(\top, [0, 3], [0, 3], \perp)$

$(\top, [0, 3], [0, 3], \perp)$

$(\top, [0, 6], [0, 3], \perp)$

$(\top, [0, 6], [0, 6], \perp)$

$(\top, [0, 9], [0, 9], \perp)$

$(\top, [0, 12], [0, 9], \perp)$

$(\top, [0, 12], [0, 9], [10, 12])$

$(\top, [0, 12], [0, 9], [10, 12])$

$$F^n(\emptyset, \emptyset, \emptyset, \emptyset) \subseteq \gamma((F^\#)^n(\perp, \perp, \perp, \perp))$$

Proof by Induction

Key Aspect of Soundness:

$$F(\gamma(a)) \subseteq \gamma(F^\#(a)) \quad (*)$$

Then if

$$F^n(\bar{\emptyset}) \subseteq \gamma((F^\#)^n(\bar{\perp}))$$

we have

$$F(F^n(\bar{\emptyset})) \subseteq^{\text{IH, monotone}} F(\gamma((F^\#)^n(\bar{\perp}))) \subseteq^{(*)} \gamma(F^\#((F^\#)^n(\bar{\perp})))$$